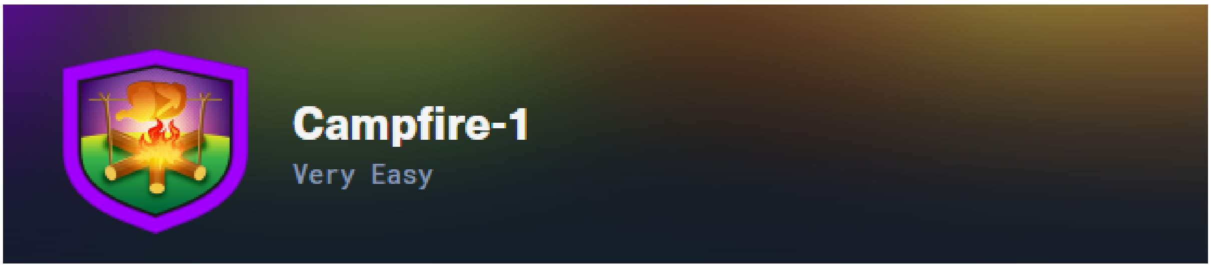


Campfire-1

Very Easy

Senario ➖

Alonzo Spotted Weird files on his computer and informed the newly assembled SOC Team. Assessing the situation it is believed a Kerberoasting attack may have occurred in the network. It is your job to confirm the findings by analyzing the provided evidence.

Analysis ➖

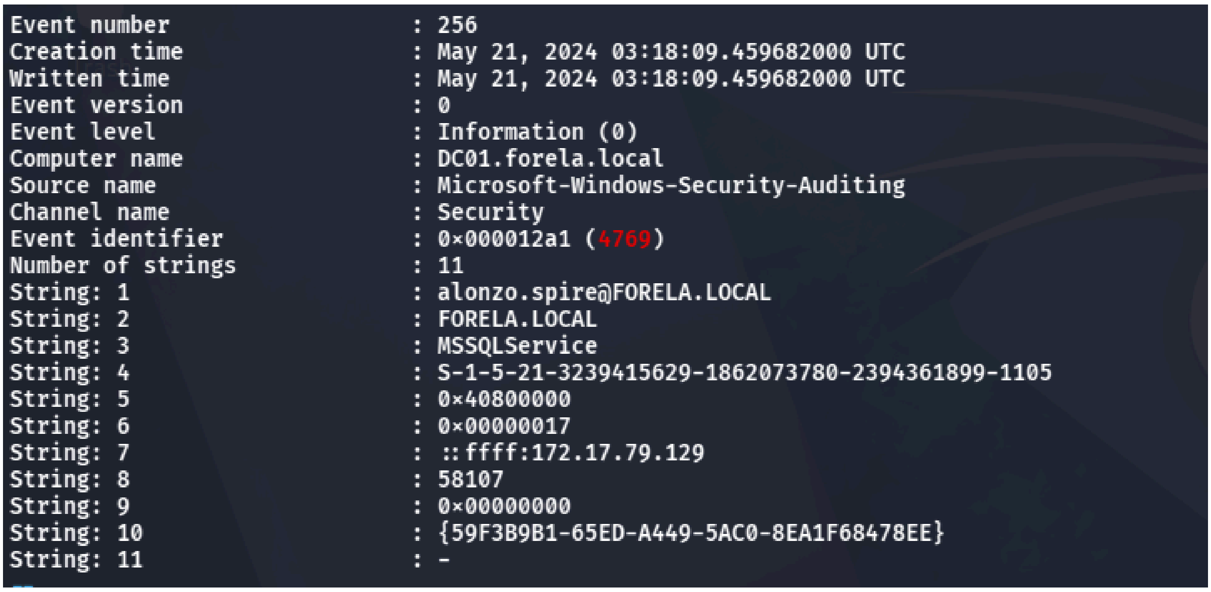
First of all Kerberoasting is when Attackers Abuse Service Accounts (Accounts set with SPNs) to Obtain their Account Hash and Crack it Offline . We have been supplied with the DC Windows Event Logs file and PowerShell Event Logs with other Preftech Files .

I Will Start Parsing the DC Event Logs into an XML file and start Analyzing

```
┌──(kali㉿kali)-[~/Htb/Campfire-1/Triage/Domain Controller]
└─$ evtxexport SECURITY-DC.evtx > Dc-EventLogs.xml
```

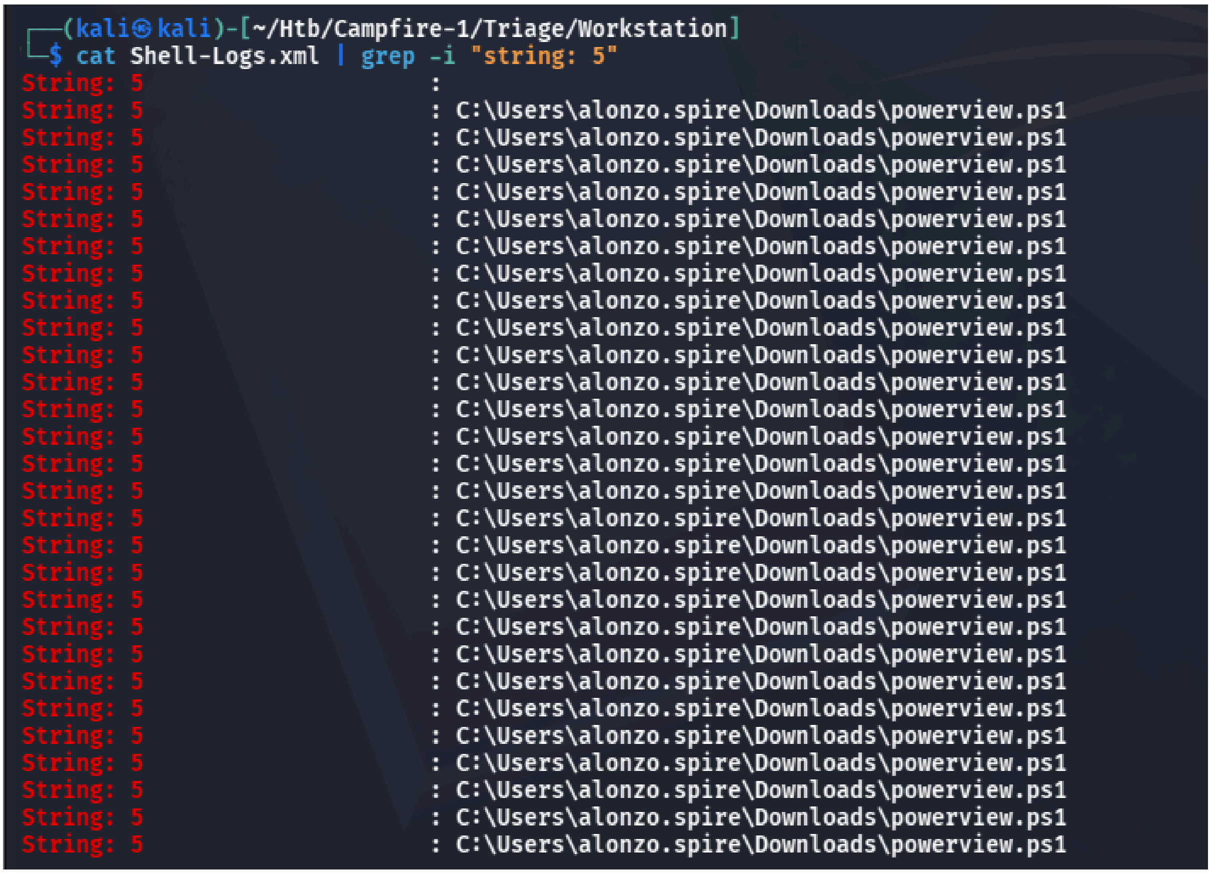
When Looking for Kerberoasting we think of the TGS request's Logs in the Domain which has the Event ID of 4769 filtering that will speedup the process of parsing the Logs to Find interesting ones out of the noise

```
┌──(kali㉿kali)-[~/Htb/Campfire-1/Triage/Domain Controller]
└─$ cat Dc-EventLogs.xml | grep 4769 -A 12 -B 8
```



```
Event number                   : 256
Creation time                  : May 21, 2024 03:18:09.459682000 UTC
Written time                   : May 21, 2024 03:18:09.459682000 UTC
Event version                  : 0
Event level                    : Information (0)
Computer name                  : DC01.forela.local
Source name                    : Microsoft-Windows-Security-Auditing
Channel name                   : Security
Event identifier               : 0×000012a1 (4769)
Number of strings              : 11
String: 1                      : alonzo.spire@FORELA.LOCAL
String: 2                      : FORELA.LOCAL
String: 3                      : MSSQLService
String: 4                      : S-1-5-21-3239415629-1862073780-2394361899-1105
String: 5                      : 0×40800000
String: 6                      : 0×00000017
String: 7                      : ::ffff:172.17.79.129
String: 8                      : 58107
String: 9                      : 0×00000000
String: 10                     : {59F3B9B1-65ED-A449-5AC0-8EA1F68478EE}
String: 11                     : -
```

We can observe this interesting TGS request upon others where the user alonzo.spire requested a TGS at **2024-05-21 03:18:09** to the **MSSQLService** Service with the IP Address of **172.17.79.129** . Now that we have initial intel about the victim service and the attacker profile used during his attack we can start analyzing the Powershell Logs to map the exact Commands being used and how the attack was Executed .

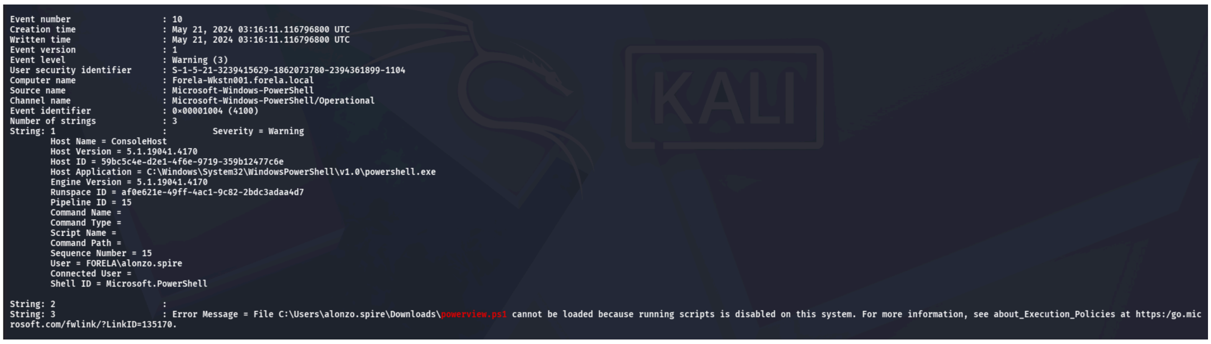


```
┌──(kali㉿kali)-[~/Htb/Campfire-1/Triage/Workstation]
└─$ cat Shell-Logs.xml | grep -i "string: 5"
String: 5                      :
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
String: 5                      : C:\Users\alonzo.spire\Downloads\powerview.ps1
```

Grepping the Scripts being Executed in the Logs we can see that the attacker mainly used **PowerView.ps1** which is a very common AD Enumeration & Attack Tool Next is to check each command one by one to see the logical steps the attacker made .

```
┌──(kali㉿kali)-[~/Htb/Campfire-1/Triage/Workstation]
└─$ cat Shell-Logs.xml | grep -i "powerview.ps1" -B 70 -m 1
```

A little explanation regards the above command , the powerview script can hold a lot of code so i used -B which is “Print Before” to print at least 70 lines before the word “powerview.ps1” and used -m which is “matching” i want to match the first Occurrence so i can connect it to a timeline where the attacker used his Tool for the first time.

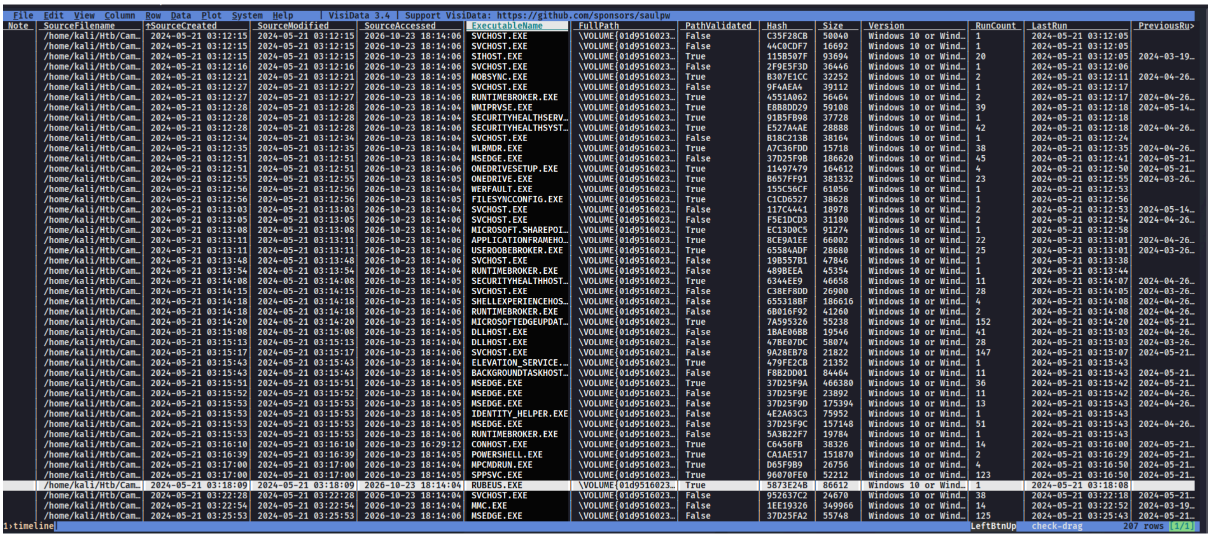


```
Event number                : 10
Creation time               : May 21, 2024 03:16:11.116796800 UTC
Written time                : May 21, 2024 03:16:11.116796800 UTC
Event version               : 1
Event level                 : Warning (3)
User security identifier    : S-1-5-21-3239415629-1862073780-2394361899-1104
Computer name               : Forela-Wkstn001.forela.local
Source name                 : Microsoft-Windows-PowerShell
Channel name                : Microsoft-Windows-PowerShell/Operational
Event identifier            : 0×00001004 (4100)
Number of strings           : 3
String: 1                   :       Severity = Warning
        Host Name = ConsoleHost
        Host Version = 5.1.19041.4170
        Host ID = 59bc5c4e-d2e1-4f6e-9719-359b12477c6e
        Host Application = C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
        Engine Version = 5.1.19041.4170
        Runspace ID = af0e621e-49ff-4ac1-9c82-2bdc3adaa4d7
        Pipeline ID = 15
        Command Name =
        Command Type =
        Script Name =
        Command Path =
        Sequence Number = 15
        User = FORELA\alonzo.spire
        Connected User =
        Shell ID = Microsoft.PowerShell

String: 2                   :
String: 3                   : Error Message = File C:\Users\alonzo.spire\Downloads\powerview.ps1 cannot be loaded because running scripts is disabled on this system. For more information, see about_Execution_Policies at https:/go.mic
rosoft.com/fwlink/?LinkID=135170.
```

We can Observe Bunch of useful informations , the first is attacker used his tool initially but it failed due to Execution Policy but it could be easily bypassed and it seems that what the attacker did going to the second time he Runned the Powerview.ps1 it ran successfully at **2024-05-21 03:16:32**

Using “pecmd” and “TimeLine Explorer” Tools to analyze the prefetch Files to see what programs been Executed during that Time



```
 File  Edit  View  Column  Row  Data  Plot  System  Help    | VisiData 3.4 | Support VisiData: https://github.com/sponsors/saulpw
Note  SourceFilename          ↑SourceCreated      SourceModified      SourceAccessed      ExecutableName       FullPath             PathValidated  Hash      Size    Version              RunCount  LastRun              PreviousRu>
      /home/kali/Htb/Cam…   2024-05-21 03:12:15 2024-05-21 03:12:15 2026-10-23 18:14:06 SVCHOST.EXE          \VOLUME{01d9516023…  False          C35F28CB  50040   Windows 10 or Wind…  1         2024-05-21 03:12:05
      /home/kali/Htb/Cam…   2024-05-21 03:12:15 2024-05-21 03:12:15 2026-10-23 18:14:06 SVCHOST.EXE          \VOLUME{01d9516023…  False          44C0CDF7  16692   Windows 10 or Wind…  1         2024-05-21 03:12:05
      /home/kali/Htb/Cam…   2024-05-21 03:12:15 2024-05-21 03:12:15 2026-10-23 18:14:06 SIHOST.EXE           \VOLUME{01d9516023…  True           115B507F  93694   Windows 10 or Wind…  20        2024-05-21 03:12:05  2024-03-19…
      /home/kali/Htb/Cam…   2024-05-21 03:12:16 2024-05-21 03:12:16 2026-10-23 18:14:06 SVCHOST.EXE          \VOLUME{01d9516023…  False          2F9E5F3D  36446   Windows 10 or Wind…  1         2024-05-21 03:12:06
      /home/kali/Htb/Cam…   2024-05-21 03:12:21 2024-05-21 03:12:21 2026-10-23 18:14:05 MOBSYNC.EXE          \VOLUME{01d9516023…  True           B307E1CC  32252   Windows 10 or Wind…  2         2024-05-21 03:12:11  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:12:27 2024-05-21 03:12:27 2026-10-23 18:14:05 SVCHOST.EXE          \VOLUME{01d9516023…  False          9F4AEA4   39112   Windows 10 or Wind…  1         2024-05-21 03:12:17
      /home/kali/Htb/Cam…   2024-05-21 03:12:27 2024-05-21 03:12:27 2026-10-23 18:14:06 RUNTIMEBROKER.EXE    \VOLUME{01d9516023…  True           4551A062  56464   Windows 10 or Wind…  2         2024-05-21 03:12:17  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:12:28 2024-05-21 03:12:28 2026-10-23 18:14:04 WMIPRVSE.EXE         \VOLUME{01d9516023…  True           E8B8DD29  59108   Windows 10 or Wind…  39        2024-05-21 03:12:18  2024-05-14…
      /home/kali/Htb/Cam…   2024-05-21 03:12:28 2024-05-21 03:12:28 2026-10-23 18:14:04 SECURITYHEALTHSERV…  \VOLUME{01d9516023…  True           91B5FB98  37728   Windows 10 or Wind…  1         2024-05-21 03:12:18
      /home/kali/Htb/Cam…   2024-05-21 03:12:28 2024-05-21 03:12:28 2026-10-23 18:14:06 SECURITYHEALTHSYST…  \VOLUME{01d9516023…  True           E527A4AE  28888   Windows 10 or Wind…  42        2024-05-21 03:12:18  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:12:34 2024-05-21 03:12:34 2026-10-23 18:14:04 SVCHOST.EXE          \VOLUME{01d9516023…  False          B18C213B  38164   Windows 10 or Wind…  1         2024-05-21 03:12:24
      /home/kali/Htb/Cam…   2024-05-21 03:12:35 2024-05-21 03:12:35 2026-10-23 18:14:04 WLRMDR.EXE           \VOLUME{01d9516023…  True           A7C36FDD  15718   Windows 10 or Wind…  38        2024-05-21 03:12:35  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:12:51 2024-05-21 03:12:51 2026-10-23 18:14:04 MSEDGE.EXE           \VOLUME{01d9516023…  False          37D25F9B  186620  Windows 10 or Wind…  45        2024-05-21 03:12:41  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:12:51 2024-05-21 03:12:51 2026-10-23 18:14:06 ONEDRIVESETUP.EXE    \VOLUME{01d9516023…  True           11497479  164612  Windows 10 or Wind…  4         2024-05-21 03:12:50  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:12:55 2024-05-21 03:12:55 2026-10-23 18:14:05 ONEDRIVE.EXE         \VOLUME{01d9516023…  True           B657FF91  381332  Windows 10 or Wind…  23        2024-05-21 03:12:55  2024-03-26…
      /home/kali/Htb/Cam…   2024-05-21 03:12:56 2024-05-21 03:12:56 2026-10-23 18:14:04 WERFAULT.EXE         \VOLUME{01d9516023…  True           155C56CF  61056   Windows 10 or Wind…  1         2024-05-21 03:12:53
      /home/kali/Htb/Cam…   2024-05-21 03:12:56 2024-05-21 03:12:56 2026-10-23 18:14:05 FILESYNCCONFIG.EXE   \VOLUME{01d9516023…  True           C1CD6527  38628   Windows 10 or Wind…  1         2024-05-21 03:12:56
      /home/kali/Htb/Cam…   2024-05-21 03:13:03 2024-05-21 03:13:03 2026-10-23 18:14:04 SVCHOST.EXE          \VOLUME{01d9516023…  False          117C4441  18978   Windows 10 or Wind…  2         2024-05-21 03:12:53  2024-05-14…
      /home/kali/Htb/Cam…   2024-05-21 03:13:05 2024-05-21 03:13:05 2026-10-23 18:14:06 SVCHOST.EXE          \VOLUME{01d9516023…  False          F5E1DCD3  31180   Windows 10 or Wind…  2         2024-05-21 03:12:54  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:13:08 2024-05-21 03:13:08 2026-10-23 18:14:04 MICROSOFT.SHAREPOI…  \VOLUME{01d9516023…  True           EC13D0C5  91274   Windows 10 or Wind…  1         2024-05-21 03:12:58
      /home/kali/Htb/Cam…   2024-05-21 03:13:11 2024-05-21 03:13:11 2026-10-23 18:14:06 APPLICATIONFRAMEHO…  \VOLUME{01d9516023…  True           8CE9A1EE  66002   Windows 10 or Wind…  22        2024-05-21 03:13:01  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:13:11 2024-05-21 03:13:11 2026-10-23 18:14:06 USEROOBEBROKER.EXE   \VOLUME{01d9516023…  True           65584ADF  28680   Windows 10 or Wind…  25        2024-05-21 03:13:01  2024-03-26…
      /home/kali/Htb/Cam…   2024-05-21 03:13:48 2024-05-21 03:13:48 2026-10-23 18:14:06 SVCHOST.EXE          \VOLUME{01d9516023…  False          19B557B1  47846   Windows 10 or Wind…  1         2024-05-21 03:13:38
      /home/kali/Htb/Cam…   2024-05-21 03:13:54 2024-05-21 03:13:54 2026-10-23 18:14:04 RUNTIMEBROKER.EXE    \VOLUME{01d9516023…  False          489BEEA   45354   Windows 10 or Wind…  1         2024-05-21 03:13:44
      /home/kali/Htb/Cam…   2024-05-21 03:14:08 2024-05-21 03:14:08 2026-10-23 18:14:05 SECURITYHEALTHHOST…  \VOLUME{01d9516023…  True           6344EE9   46658   Windows 10 or Wind…  11        2024-05-21 03:14:07  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:14:15 2024-05-21 03:14:15 2026-10-23 18:14:04 SVCHOST.EXE          \VOLUME{01d9516023…  False          C38EF8DD  26900   Windows 10 or Wind…  28        2024-05-21 03:14:05  2024-03-26…
      /home/kali/Htb/Cam…   2024-05-21 03:14:18 2024-05-21 03:14:18 2026-10-23 18:14:05 SHELLEXPERIENCEHOS…  \VOLUME{01d9516023…  False          655318BF  186616  Windows 10 or Wind…  4         2024-05-21 03:14:08  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:14:18 2024-05-21 03:14:18 2026-10-23 18:14:06 RUNTIMEBROKER.EXE    \VOLUME{01d9516023…  False          6B016F92  41260   Windows 10 or Wind…  2         2024-05-21 03:14:08  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:14:20 2024-05-21 03:14:20 2026-10-23 18:14:05 MICROSOFTEDGEUPDAT…  \VOLUME{01d9516023…  True           7A595326  55238   Windows 10 or Wind…  152       2024-05-21 03:14:20  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:15:08 2024-05-21 03:15:08 2026-10-23 18:14:05 DLLHOST.EXE          \VOLUME{01d9516023…  False          1BAE06BB  19546   Windows 10 or Wind…  41        2024-05-21 03:15:03  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:15:13 2024-05-21 03:15:13 2026-10-23 18:14:04 DLLHOST.EXE          \VOLUME{01d9516023…  False          47BE07DC  58074   Windows 10 or Wind…  28        2024-05-21 03:15:03  2024-03-26…
      /home/kali/Htb/Cam…   2024-05-21 03:15:17 2024-05-21 03:15:17 2026-10-23 18:14:06 SVCHOST.EXE          \VOLUME{01d9516023…  False          9A28EB78  21822   Windows 10 or Wind…  147       2024-05-21 03:15:07  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:15:43 2024-05-21 03:15:43 2026-10-23 18:14:04 ELEVATION_SERVICE.…  \VOLUME{01d9516023…  True           479FE2CB  21352   Windows 10 or Wind…  1         2024-05-21 03:15:43
      /home/kali/Htb/Cam…   2024-05-21 03:15:43 2024-05-21 03:15:43 2026-10-23 18:14:05 BACKGROUNDTASKHOST…  \VOLUME{01d9516023…  False          F8B2DD01  84464   Windows 10 or Wind…  11        2024-05-21 03:15:43  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:15:51 2024-05-21 03:15:51 2026-10-23 18:14:05 MSEDGE.EXE           \VOLUME{01d9516023…  True           37D25F9A  466380  Windows 10 or Wind…  36        2024-05-21 03:15:42  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:15:52 2024-05-21 03:15:52 2026-10-23 18:14:04 MSEDGE.EXE           \VOLUME{01d9516023…  False          37D25F9E  23892   Windows 10 or Wind…  11        2024-05-21 03:15:42  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:15:53 2024-05-21 03:15:53 2026-10-23 18:14:05 MSEDGE.EXE           \VOLUME{01d9516023…  False          37D25F9D  175394  Windows 10 or Wind…  13        2024-05-21 03:15:43  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:15:53 2024-05-21 03:15:53 2026-10-23 18:14:05 IDENTITY_HELPER.EXE  \VOLUME{01d9516023…  False          4E2A63C3  75952   Windows 10 or Wind…  1         2024-05-21 03:15:43
      /home/kali/Htb/Cam…   2024-05-21 03:15:53 2024-05-21 03:15:53 2026-10-23 18:14:05 MSEDGE.EXE           \VOLUME{01d9516023…  False          37D25F9C  157148  Windows 10 or Wind…  51        2024-05-21 03:15:43  2024-04-26…
      /home/kali/Htb/Cam…   2024-05-21 03:15:53 2024-05-21 03:15:53 2026-10-23 18:14:06 RUNTIMEBROKER.EXE    \VOLUME{01d9516023…  False          5A3B22F7  19784   Windows 10 or Wind…  1         2024-05-21 03:15:43
      /home/kali/Htb/Cam…   2024-05-21 03:16:10 2024-05-21 03:16:10 2026-10-23 16:29:12 CONHOST.EXE          \VOLUME{01d9516023…  True           C6456FB   38326   Windows 10 or Wind…  14        2024-05-21 03:16:00  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:16:39 2024-05-21 03:16:39 2026-10-23 18:14:05 POWERSHELL.EXE       \VOLUME{01d9516023…  True           CA1AE517  151870  Windows 10 or Wind…  2         2024-05-21 03:16:29  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:17:00 2024-05-21 03:17:00 2026-10-23 18:14:04 MPCMDRUN.EXE         \VOLUME{01d9516023…  True           D65F9B9   26756   Windows 10 or Wind…  4         2024-05-21 03:16:50  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:17:00 2024-05-21 03:17:00 2026-10-23 18:14:05 SPPSVC.EXE           \VOLUME{01d9516023…  True           96070FE0  52212   Windows 10 or Wind…  123       2024-05-21 03:16:50  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:18:09 2024-05-21 03:18:09 2026-10-23 18:14:04 RUBEUS.EXE           \VOLUME{01d9516023…  True           5873E24B  86612   Windows 10 or Wind…  1         2024-05-21 03:18:08
      /home/kali/Htb/Cam…   2024-05-21 03:22:28 2024-05-21 03:22:28 2026-10-23 18:14:04 SVCHOST.EXE          \VOLUME{01d9516023…  False          952637C2  24670   Windows 10 or Wind…  38        2024-05-21 03:22:18  2024-05-21…
      /home/kali/Htb/Cam…   2024-05-21 03:22:54 2024-05-21 03:22:54 2026-10-23 18:14:04 MMC.EXE              \VOLUME{01d9516023…  False          1EE19326  349966  Windows 10 or Wind…  14        2024-05-21 03:22:52  2024-03-19…
      /home/kali/Htb/Cam…   2024-05-21 03:25:53 2024-05-21 03:25:53 2026-10-23 18:14:06 MSEDGE.EXE           \VOLUME{01d9516023…  False          37D25FA2  55748   Windows 10 or Wind…  125       2024-05-21 03:25:43  2024-05-21…
1›timeline                                                                                                                                              LeftBtnUp  check-drag          207 rows [1/1]
```

Around the Time of Executing Powerview.ps1 we can Notice Rubeues.exe been used Too which is Ticket manipulation Tool that allows the creation and the impersonation of users via TGT and TGS most likely its the tool been used to obtain the TGS of the Service as it capable of requesting TGS by using the Kerberos mode the tool offers which automatically find a user setted with Service Principal

Name(SPN) and try to obtain its hash via TGS request. We can see its full path is **C:\Users\Alonzo.spire\Downloads\Rubeus.exe** . we also notice it is been Executed only one time and it was initially Runned at **2024-05-21 03:18:08**

➖Remediation

To mitigate these risks, implement Windows Defender Application Control (WDAC) or AppLocker to enforce PowerShell Constrained Language Mode (CLM), which strictly blocks unauthorized scripts and overrides simple execution policy bypass flags. To neutralize the Kerberoasting threat, migrate vulnerable service accounts to Group Managed Service Accounts (gMSAs) to leverage auto-rotating, 120-character passwords, or enforce a strict 25+ character password requirement for legacy accounts to make offline brute-forcing impossible. Finally, ensure all service accounts follow the principle of least privilege to limit the blast radius .